From Curvature to Logistic Regression

ES 335 / ES 678 Machine Learning, Sem I 2026 — Prof. Udit Bhatia Source: `Lectures/L06_logistic_regression_1.pdf` (37 slides) · Previous: Normal equation and PCA · Next: Ridge, lasso and elastic net

> **How to use this file.** Part A builds the *new* maths Lecture 6 needs, on top of Lectures 1–5: the **Hessian** and the **curvature test** for many variables, then **probability → odds → log-odds → sigmoid**, the **Bernoulli likelihood**, the **vector chain rule** and **Newton's method**. Part B goes through all 37 slides. Every derivation is expanded line by line and checked with numbers on a tiny 6-road dataset that runs through the whole note. Part C is practice with answers.

> **What this lecture is really about:** the linear-regression recipe from L3–L5, re-run for a **yes/no** outcome.

$$\underbrace{\Pr(Y=1 \mid x)=\sigma(x^\top\theta)}_{\text{probability model}} \xrightarrow{-\log\ \text{likelihood}} \underbrace{J(\theta)=\sum_i \left[\log(1+e^{z_i})-y_i z_i\right]}_{\text{binary cross-entropy}} \xrightarrow{\nabla,\ \nabla^2} \underbrace{X^\top(p-y),\quad X^\top W}_{\text{gradient, convex Hes}}$$

The first half of the lecture builds the tool (**curvature in many directions = eigenvalues of the Hessian**). The second half builds the model and then uses the tool to prove that its loss is **convex**: one valley, no false minima.

Step-up path

You need. Gradients, eigenvalues, and the likelihood recipe.

1. Read curvature from a second derivative, and then from a Hessian.
2. Approximate a surface by a quadratic, and jump to its bottom: that is Newton.
3. Pass from probability to odds to log-odds, the scale where a linear score makes sense.
4. Apply the likelihood recipe to a coin at each x, which gives cross-entropy.
5. Show the Hessian is positive semidefinite, so logistic loss is convex.

You can stop when you can write the gradient and Hessian of logistic loss, and say what happens on separable data.

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Part B — Slides · All 37 slides

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Part A Maths From the Ground Up

A1 Curvature in one variable

For $f(t)$: the **first derivative** $f'(t)$ is the slope; the **second derivative** $f''(t)$ is how fast the slope itself changes.

$f''(t)$	Shape	Example
> 0	bends **up** (a cup ∪): slope increasing	t^2: $f'' = 2$
$= 0$	straight line	$3t + 1$: $f'' = 0$
< 0	bends **down** (a cap ∩)	$-t^2$: $f'' = -2$

For a parabola $f(t) = at^2 + bt + c$: $f' = 2at + b$, $f'' = 2a$. The sign of a is the whole story.

Curvature test (1-D). A twice-differentiable f is **convex** (every chord lies above the curve, L2 §A7) **iff** $f''(t) \geq 0$ for every t.

Why it matters: at a convex function's flat point ($f' = 0$) you are at the **global** minimum. No other valley can exist.

A2 Gradient and Hessian

With $\theta = (\theta_1, \ldots, \theta_d)$ there are d first partials and $d \times d$ second partials.

$$\nabla J(\theta) = \begin{bmatrix} \partial J/\partial \theta_1 \\ \vdots \\ \partial J/\partial \theta_d \end{bmatrix} \ (d \times 1), \qquad H(\theta) = \nabla^2 J(\theta) = \begin{bmatrix} \dfrac{\partial^2 J}{\partial \theta_1^2} & \cdots & \dfrac{\partial^2 J}{\partial \theta_1 \partial \theta_d} \\ \vdots & \ddots & \vdots \\ \dfrac{\partial^2 J}{\partial \theta_d \partial \theta_1} & \cdots & \dfrac{\partial^2 J}{\partial \theta_d^2} \end{bmatrix} \ (d \times d).$$

Reading the Hessian.

- **Diagonal** H_{jj}: ordinary 1-D curvature along axis j (hold all other θ's fixed).
- **Off-diagonal** H_{jk}: how moving θ_k changes the slope in the θ_j direction ("coupling").
- If second partials are continuous, the order doesn't matter: $\partial^2 J/\partial\theta_j\partial\theta_k = \partial^2 J/\partial\theta_k\partial\theta_j$ → H **is symmetric** (Schwarz/Clairaut). That's what lets us use the spectral theorem from L5.

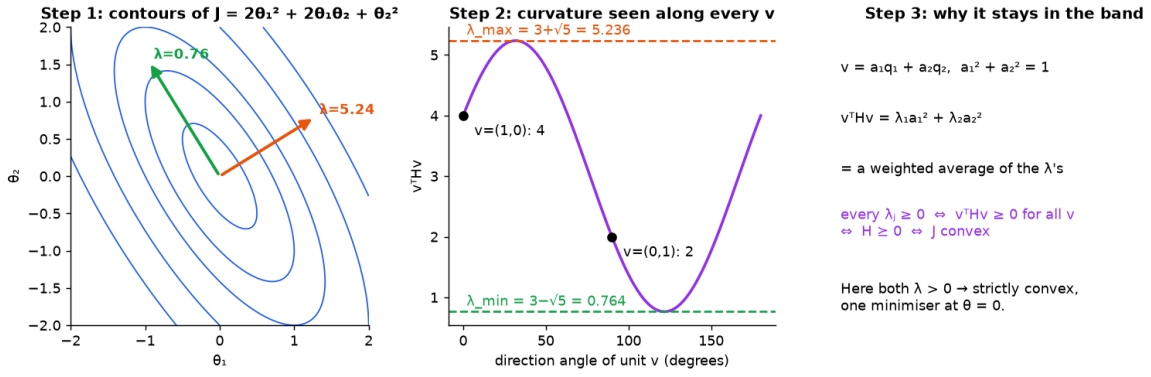
Worked example (slide 7). $J(\theta_1, \theta_2) = 2\theta_1^2 + 2\theta_1\theta_2 + \theta_2^2$.

Step	Result
$\partial J / \partial \theta_1$	$4\theta_1 + 2\theta_2$
$\partial J / \partial \theta_2$	$2\theta_1 + 2\theta_2$
$\partial^2 J / \partial \theta_1^2$ (differentiate row 1 by θ_1)	4
$\partial^2 J / \partial \theta_1 \partial \theta_2$ (row 1 by θ_2)	2
$\partial^2 J / \partial \theta_2 \partial \theta_1$ (row 2 by θ_1)	2 ✓ same
$\partial^2 J / \partial \theta_2^2$	2

$$H = \begin{bmatrix} 4 & 2 \\ 2 & 2 \end{bmatrix} \quad (\text{constant, because } J \text{ is quadratic}).$$

Shortcut for quadratics. If $J(\theta) = \frac{1}{2}\theta^\top A\theta + b^\top\theta + c$ with A symmetric, then $\nabla J = A\theta + b$ and $H = A$. Here $J = \frac{1}{2}\theta^\top \begin{bmatrix} 4 & 2 \\ 2 & 2 \end{bmatrix} \theta$ ✓. Least squares $J = \frac{1}{2}\|y - X\theta\|^2$ has $H = X^\top X$ (L5).

A3 Slicing a surface along a direction



Contours of J = 2θ₁² + 2θ₁θ₂ + θ₂², the curvature vᵀHv for every unit direction, and why it is trapped between the two eigenvalues.

A multivariable function is hard to "see". So **cut it with a line** and look at the 1-D slice:

$$\theta(t) = \theta + tv \;\; (v \text{ a fixed unit direction}), \qquad g(t) = J(\theta + tv).$$

g is an ordinary function of one number t, so A1 applies.

First derivative (chain rule; each coordinate $\theta_k + tv_k$ has t-derivative v_k):

$$g'(t) = \sum_{k=1}^{d} \frac{\partial J}{\partial \theta_k}(\theta + tv)\, v_k = \nabla J(\theta + tv)^\top v.$$

Second derivative (differentiate each $\partial J/\partial \theta_k$ again, same chain rule):

$$g''(t) = \sum_{j=1}^{d}\sum_{k=1}^{d} v_j \frac{\partial^2 J}{\partial \theta_j \partial \theta_k}(\theta + tv)\, v_k = v^\top H(\theta + tv)\, v.$$

At $t = 0$: $g''(0) = v^\top H v$ **is the curvature you feel walking through θ in direction v.**

Numbers (H from A2).

- $v = (1, 0)$: $v^\top H v = H_{11} = 4$.
- $v = (0, 1)$: $v^\top H v = H_{22} = 2$.
- $v = \frac{1}{\sqrt{2}}(1, 1)$: $\frac{1}{2}(4 + 2 + 2 + 2) = 5$.
- $v = \frac{1}{\sqrt{2}}(1, -1)$: $\frac{1}{2}(4 - 2 - 2 + 2) = 1$.

Different directions, different curvatures. Which is the biggest? Which is the smallest? → A5.

A4 Taylor's quadratic approximation

1-D Taylor at $t = 0$: $g(t) \approx g(0) + tg'(0) + \frac{1}{2}t^2 g''(0)$. Substitute $g'(0) = \nabla J^\top v$, $g''(0) = v^\top H v$ and write the step as $\Delta = tv$:

$$J(\theta + \Delta) \approx J(\theta) + \underbrace{\nabla J(\theta)^\top \Delta}_{\text{first order: tilt}} + \underbrace{\tfrac{1}{2}\Delta^\top H(\theta)\Delta}_{\text{second order: bend}}.$$

- $\nabla J^\top \Delta$ says whether J starts going up or down along Δ.
- $\Delta^\top H \Delta$ is a **number** (not the vector $H\Delta$): the bend along Δ.
- At a stationary point $\nabla J = 0$, only the bend is left: $J(\theta + \Delta) - J(\theta) \approx \frac{1}{2}\Delta^\top H \Delta$. If that is ≥ 0 for every Δ, you're at a minimum.
- For a quadratic J, the approximation is **exact** (L5 slide 23).

A5 The eigenvalue test for curvature

H is symmetric, so (spectral theorem, L5 §A6) $H = Q\Lambda Q^\top$ with orthonormal eigenvectors q_j ($Hq_j = \lambda_j q_j$).

Step 1 — write v in eigen-coordinates. $a = Q^\top v$ ($a_j = q_j^\top v$, the shadow of v on q_j), and $v = Qa = \sum_j a_j q_j$. Since $Q^\top Q = I$, $\|a\| = \|v\| = 1$, i.e. $\sum_j a_j^2 = 1$.

Step 2 — substitute.

$$v^\top H v = (Qa)^\top (Q\Lambda Q^\top)(Qa) = a^\top \underbrace{Q^\top Q}_{I} \Lambda \underbrace{Q^\top Q}_{I} a = a^\top \Lambda a = \sum_{j=1}^{d} \lambda_j a_j^2.$$

Step 3 — read it. Weights $a_j^2 \geq 0$ sum to 1, so $v^\top H v$ is a **weighted average of the eigenvalues**:

$$\lambda_{\min} \leq v^\top H v \leq \lambda_{\max}, \qquad \text{with equality at } v = q_{\min},\ q_{\max}.$$

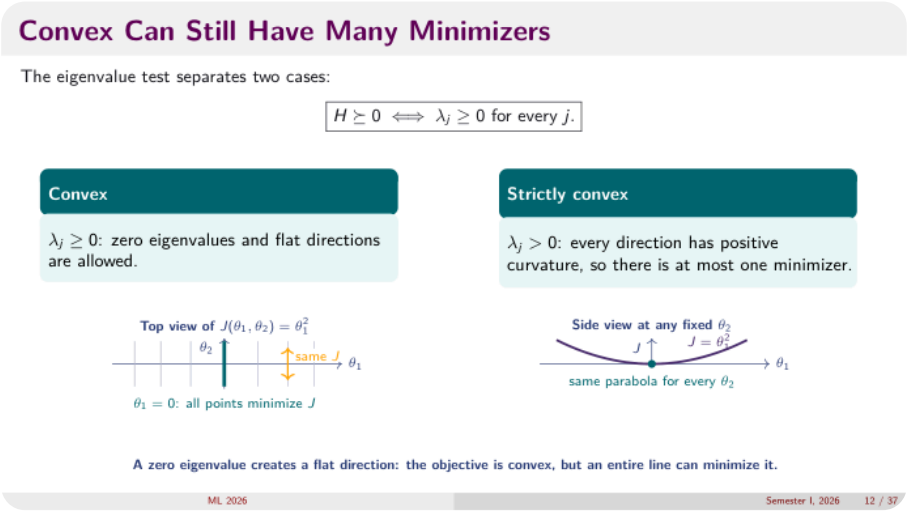
Step 4 — the test.

$$v^\top H v \geq 0 \text{ for all } v \iff \lambda_j \geq 0 \text{ for all } j \iff H \succeq 0 \ (\text{PSD}).$$

(⇐: an average of non-negative numbers is non-negative. ⇒: pick $v = q_j$, then $v^\top H v = \lambda_j$, so each λ_j must be ≥ 0.)

Numbers. $H = \begin{bmatrix} 4 & 2 \\ 2 & 2 \end{bmatrix}$: trace $T = 6$, det $D = 8 - 4 = 4$, so $\lambda = \frac{T}{2} \pm \sqrt{\frac{T^2}{4} - D} = 3 \pm \sqrt{5} = 5.236,\ 0.764$. The four curvatures from A3 (4, 2, 5, 1) all lie in $[0.764,\ 5.236]$ ✓. Both $\lambda > 0$ → J is strictly convex.

A6 Convex, strictly convex, unique, finite



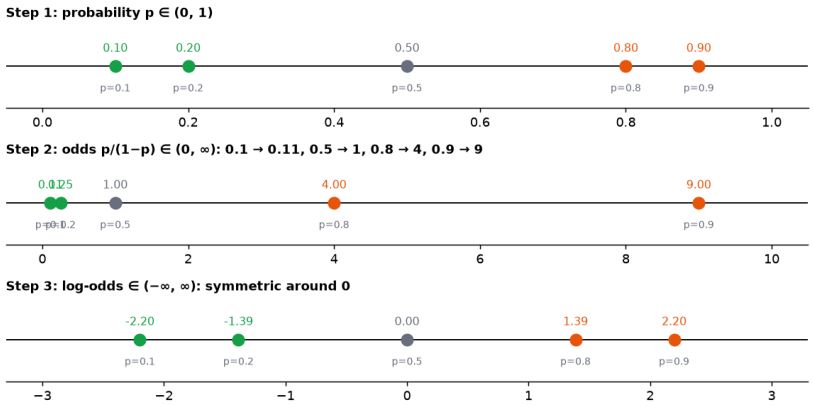
Slide 12 — $J = \theta_1^2$ is convex but the whole line $\theta_1 = 0$ minimises it: a zero eigenvalue makes a flat direction.

Four different questions people often blur:

Question	Condition	Example
Convex? (no false valleys)	$H \succeq 0$ everywhere: all $\lambda_j \geq 0$	$J = \theta_1^2$ in 2-D: $H = \mathrm{diag}(2, 0)$
Strictly convex? (at most one minimiser)	$H \succ 0$: all $\lambda_j > 0$	$J = \theta_1^2 + \theta_2^2$
Unique minimiser?	strictly convex *and* a minimiser exists	—
Finite minimiser exists?	not guaranteed even if strictly convex!	$J(\theta) = e^{-\theta}$: $J'' > 0$ but $\inf = 0$ only as $\theta \to \infty$

So: **zero eigenvalue ⇒ flat direction ⇒ possibly a whole line of minimisers** (slide 12). **Strict convexity ⇒ at most one** minimiser, but it might sit "at infinity" (logistic regression under **complete separation**, slide 33).

A7 Probability, odds, log-odds and the sigmoid



Three number lines: probabilities 0.1…0.9 → odds 0.11…9 → log-odds −2.2…2.2. Each step widens the range.

Problem: a linear score $z = x^\top \theta$ can be any real number, but a probability must be in $(0, 1)$. We need a map between them. Build it in three steps.

Step 1 — odds. Compare event to non-event:

$$\text{odds} = \frac{p}{1-p} \in (0, \infty).$$

$p = 0.8$ → odds $= 0.8/0.2 = 4$ ("4 floods for every 1 non-flood"). $p = 0.5$ → 1. $p = 0.1$ → $1/9 \approx 0.111$.

Step 2 — log-odds (the "logit").

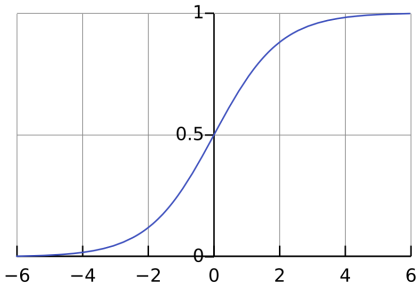
$$\text{logit}(p) = \log \frac{p}{1-p} \in (-\infty, \infty).$$

$p = 0.5 \to 0$; $p = 0.9 \to \log 9 = 2.197$; $p = 0.1 \to \log(1/9) = -2.197$. **Swapping event and non-event only flips the sign** ($\log \frac{1-p}{p} = -\log \frac{p}{1-p}$).

Step 3 — set logit = linear score, then solve for p:

Line	Operation
$\log \frac{p}{1-p} = z$	the modelling choice
$\frac{p}{1-p} = e^z$	exponentiate both sides
$p = e^z(1-p) = e^z - e^z p$	multiply out
$p + e^z p = e^z \;\Rightarrow\; p(1+e^z) = e^z$	collect p
$p = \frac{e^z}{1+e^z} = \frac{1}{1+e^{-z}} =: \sigma(z)$	divide top and bottom by e^z

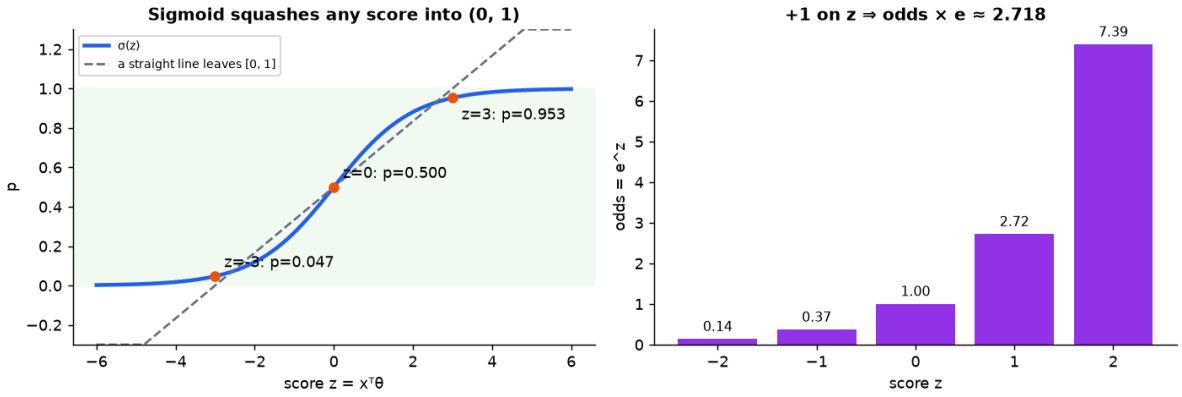
The sigmoid is just the inverse of the logit. Nothing decorative.



The logistic (sigmoid) curve σ(z) = 1/(1+e^−z). (Wikimedia Commons: Logistic-curve.svg, Public domain)

Sigmoid facts to know cold.

Fact	Why
$\sigma(0) = \frac{1}{2}$	$1/(1+1)$
$\sigma(-z) = 1 - \sigma(z)$	$\frac{1}{1+e^z} = \frac{e^{-z}}{e^{-z}+1} = 1 - \frac{1}{1+e^{-z}}$
$\sigma(3) \approx 0.953$, $\sigma(-3) \approx 0.047$	$1/(1+e^{-3}) = 1/1.0498$
$+1$ on z ⇒ odds $\times e \approx 2.718$	odds $= e^z$, and $e^{z+1} = e \cdot e^z$
$\sigma'(z) = \sigma(z)(1-\sigma(z))$	slide 29
$1 - \sigma(z) = \frac{1}{1+e^z}$	used in slide 26



Left: sigmoid vs a straight line. Right: each extra unit of score multiplies the odds by e.

Coefficient interpretation. If $z = \theta_0 + \theta_1 x_1 + \ldots$, raising x_1 by 1 adds θ_1 to the log-odds, i.e. **multiplies the odds by** e^{θ_1}. It does *not* add a fixed amount to p (the effect on p is biggest near $p = \frac{1}{2}$).

A8 The Bernoulli distribution and its likelihood

A **Bernoulli(p)** variable takes value 1 with probability p, and 0 with probability $1 - p$ (L3). One formula covers both:

$$\Pr(Y = y) = p^y (1-p)^{1-y}, \qquad y \in \{0, 1\}.$$

Check: $y = 1$: $p^1(1-p)^0 = p$ ✓. $y = 0$: $p^0(1-p)^1 = 1 - p$ ✓. The exponent acts as a **switch**.

Dataset likelihood. If observations are **conditionally independent given their** x_i, multiply:

$$L(\theta) = \prod_{i=1}^{n} p_i^{y_i} (1 - p_i)^{1 - y_i}, \qquad p_i = \sigma(x_i^\top \theta).$$

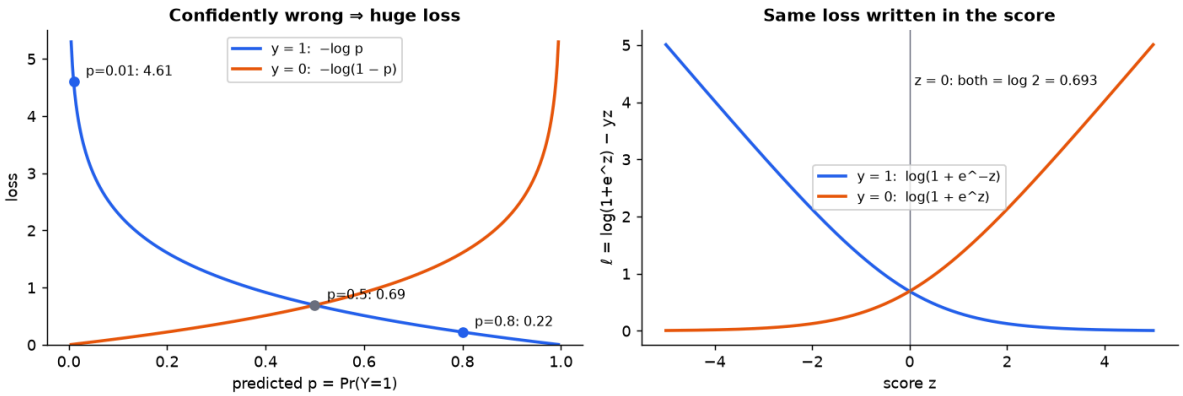
Log rules used (L1 §10): $\log \prod a_i = \sum \log a_i$ and $\log a^b = b \log a$. So

$$\log L(\theta) = \sum_{i=1}^{n} \big[y_i \log p_i + (1 - y_i) \log(1 - p_i) \big].$$

Logs turn a product of tiny numbers (underflow risk) into a sum, and don't move the argmax (log is increasing).

Careful. "Conditionally independent observations" is about **rows** (road 1's outcome doesn't affect road 2's, once you know both x's). It says nothing about **columns** (rain and slope may be correlated).

A9 Binary cross-entropy (log loss)



Left: –log p explodes as p→0 when y = 1 (p = 0.01 costs 4.61). Right: the same loss as a function of the score z.

Maximising $\log L$ = minimising its negative:

$$J(\theta) = -\sum_i \left[y_i \log p_i + (1 - y_i) \log(1 - p_i) \right].$$

Three names, one thing: **Bernoulli negative log-likelihood = binary cross-entropy = logistic loss.**

Per-example loss ℓ_i: if $y_i = 1$ it is $-\log p_i$; if $y_i = 0$ it is $-\log(1 - p_i)$.

y	p	loss	comment
1	0.8	$-\log 0.8 = 0.223$	fairly confident, right → small
1	0.5	$\log 2 = 0.693$	shrug
1	0.01	$-\log 0.01 = 4.605$	**confidently wrong → huge**
0	0.01	$-\log 0.99 = 0.010$	confidently right → ~0

In terms of the score z_i (slides 26–27): $\ell_i = \log(1 + e^{z_i}) - y_i z_i$. For $y = 1$ this is $\log(1 + e^{-z})$; for $y = 0$ it is $\log(1 + e^{z})$. Both are smooth, convex in z, and approach a straight line for large wrong-side scores (unlike the squared loss, which would grow quadratically).

Why not squared error on p? $(y-\sigma(z))^2$ is **not convex** in θ and its gradient vanishes when σ saturates, so a confidently wrong model barely gets corrected. Log loss comes from the likelihood and is convex (slide 33).

A10 The chain rule with a vector parameter

ℓ_i depends on θ only through the scalar $z_i = x_i^\top \theta$. So differentiate in two hops:

$$\nabla_\theta \ell_i = \underbrace{\frac{d\ell_i}{dz_i}}_{\text{scalar}} \cdot \underbrace{\nabla_\theta z_i}_{=x_i}.$$

Why $\nabla_\theta(x_i^\top\theta) = x_i$: $x_i^\top\theta = \sum_k x_{ik}\theta_k$, and $\partial/\partial\theta_k$ of that is x_{ik}.

Second derivative of a product $s(\theta)\,x_i$ (a scalar function times a fixed vector): $\nabla_\theta\big[s(\theta)x_i\big] = x_i\,(\nabla_\theta s)^\top$, a $d\times d$ matrix (column x_i times row $\nabla s^\top$).

A11 Outer products and $X^\top W X$

The **outer product** $xx^\top$ of a column $x\in\mathbb{R}^d$ is the $d\times d$ matrix with entries $x_j x_k$. Example: $x=(1,3)$: $xx^\top = \begin{bmatrix}1 & 3\\ 3 & 9\end{bmatrix}$. It is symmetric PSD: $v^\top xx^\top v = (x^\top v)^2 \geq 0$.

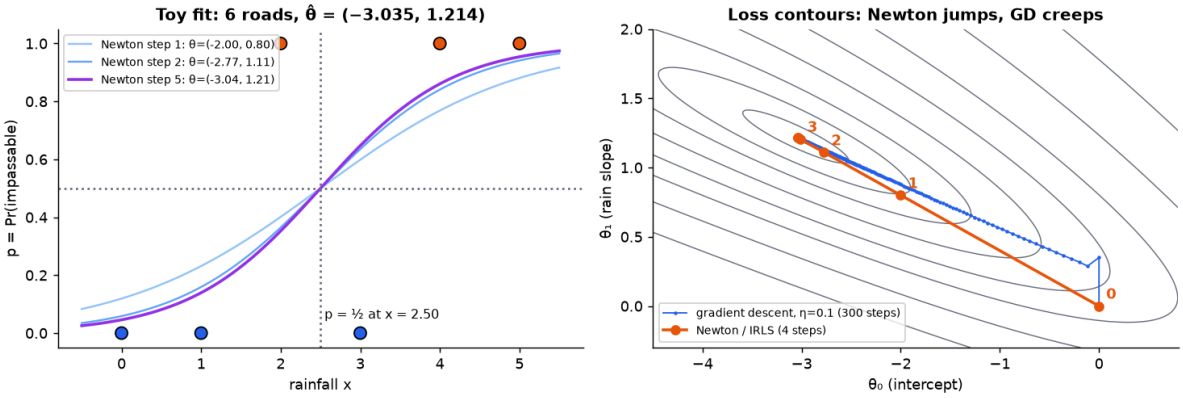
Sum of weighted outer products = one matrix product. With rows $x_i^\top$ stacked in X and $W = \mathrm{diag}(w_1..w_n)$:

$$\sum_{i=1}^{n} w_i x_i x_i^\top = X^\top W X.$$

(Entry (j,k) on both sides is $\sum_i w_i x_{ij} x_{ik}$.) With all $w_i = 1$ this is the Gram matrix $X^\top X$ of L5.

PSD check: $v^\top X^\top W X v = \sum_i w_i (x_i^\top v)^2 \geq 0$ whenever every $w_i \geq 0$.

A12 Gradient descent vs Newton (IRLS)



Toy 6-road data. Left: the fitted sigmoid after Newton steps 1, 2, 5. Right: on the loss contours, Newton reaches $\hat{\theta}$ in ~4 steps; gradient descent ($\eta = 0.1$) is still creeping after 300.

Both solve $\nabla J(\theta) = 0$ iteratively.

	Gradient descent	Newton's method
update	$\theta_{t+1} = \theta_t - \eta \nabla J(\theta_t)$	$\theta_{t+1} = \theta_t - H_t^{-1} \nabla J(\theta_t)$
uses	slope only + a step size η you choose	slope **and** curvature
idea	walk downhill a little	jump to the bottom of the local Taylor parabola (A4)
cost per step	cheap ($O(nd)$)	solve a $d \times d$ system ($O(nd^2 + d^3)$)
steps needed	many, esp. in long thin valleys (L5 A7)	few (quadratic convergence near the optimum)

Where Newton comes from. Minimise the Taylor model $J + \nabla J^\top \Delta + \frac{1}{2} \Delta^\top H \Delta$ over Δ: gradient $\nabla J + H\Delta = 0 \Rightarrow \Delta = -H^{-1} \nabla J$.

For logistic regression Newton is also called **IRLS** (iteratively reweighted least squares): each step solves a weighted least-squares problem with weights W_t.

Part B Slide by Slide (all 37 slides)

Running example used below (added here; the slides are symbolic). Six road segments, one feature (rainfall x), with intercept: $x_i^\top = (1, x_i)$, $\theta = (\theta_0, \theta_1)$.

road i	1	2	3	4	5	6
rainfall x_i	0	1	2	3	4	5
impassable y_i	0	0	1	0	1	1

Final fit (from Newton, slide 34): $\hat{\theta} = (-3.035,\ 1.214)$, fitted $p = (0.046, 0.139, 0.353, 0.647, 0.861, 0.954)$, $J(\hat{\theta}) = 2.478$.

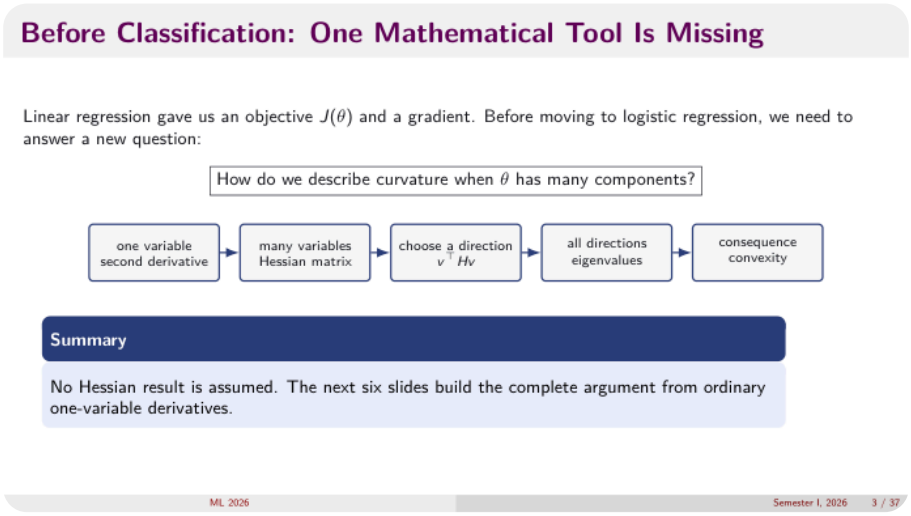
Slide 1 Title

Lecture 6: From Curvature to Logistic Regression. Prof. Udit Bhatia, Civil Engineering; Computer Science and Engineering, Semester I 2026.

Slide 2 Course Roadmap: You Are Here

Still in **L3–L6 estimation and linear models** (last lecture of the block). L5 connected the normal equation, eigen-directions and PCA. Today: **(1)** build the Hessian and the curvature test from ordinary derivatives; **(2)** derive logistic regression from probability to loss. Next block (L7–L8): trust and evaluation.

Slide 3 Before Classification: One Mathematical Tool Is Missing



Slide 3 — The tool chain: second derivative → Hessian → choose a direction $v^\top Hv$ → all directions = eigenvalues → convexity.

Linear regression gave us $J(\theta)$ and a gradient. New question: **how do we describe curvature when θ has many components?**

one variable → second derivative · many variables → **Hessian matrix** · choose a direction → $v^\top Hv$ · all directions → **eigenvalues** · consequence → **convexity**.

Summary: no Hessian result is assumed; slides 6–11 build it from 1-D derivatives (A1–A5).

Slide 4 Learning Objectives

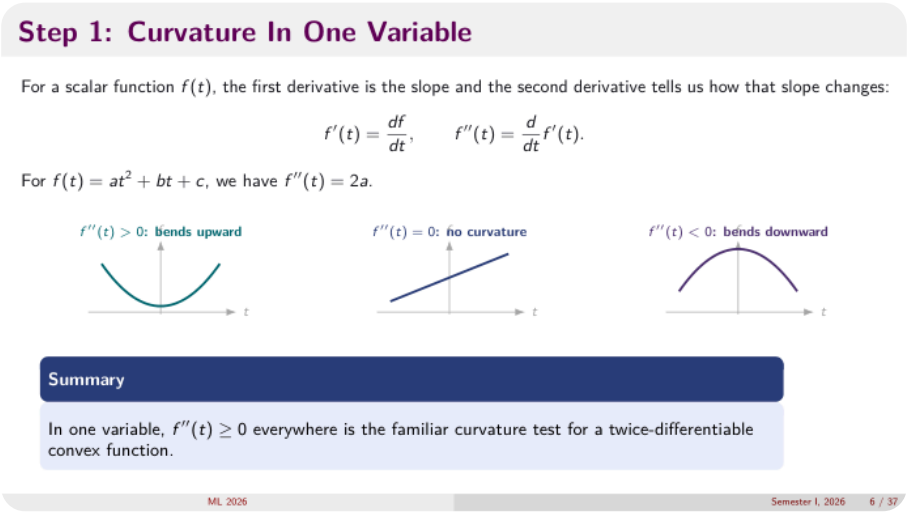
1. connect directional curvature $v^\top Hv$ with the eigenvalues of H (A5);
2. distinguish convexity, strict convexity, and uniqueness (A6);
3. derive $p \to$ odds $\to$ log-odds $\to$ sigmoid without a memorised jump (A7);
4. derive binary cross-entropy from the Bernoulli likelihood (A8–A9);
5. derive $\nabla J = X^\top(p - y)$ and $\nabla^2 J = X^\top W X$ (A10–A11);
6. prove logistic loss is convex and state when the minimiser is unique;
7. separate **probability estimation** from the final **decision threshold**.

Slide 5 Notation: Observed, Chosen, And Computed Objects

Object	Shape	Meaning	Status
$X = [x_1^\top; \ldots; x_n^\top]$	$n \times d$	design matrix	observed
$y_i \in \{0, 1\}$	scalar	binary outcome	observed
θ	$d \times 1$	model parameters	learned
$z_i = x_i^\top \theta$	scalar	unrestricted linear score	computed
$p_i = \Pr(Y_i = 1 \mid x_i)$	scalar in $(0, 1)$	event probability	modelled
$p = (p_1..p_n)^\top$	$n \times 1$	fitted probabilities	computed
$W = \mathrm{diag}(p_i(1 - p_i))$	$n \times n$	curvature weights	computed
$H = \nabla^2 J(\theta)$	$d \times d$	parameter-space curvature	computed

Summary: y_i is the observed **label**, z_i an internal **score**, p_i a modelled **probability**. Three different objects. (Toy road 4: $y_4 = 0$, $z_4 = -3.035 + 1.214 \cdot 3 = 0.607$, $p_4 = \sigma(0.607) = 0.647$.)

Slide 6 Step 1: Curvature In One Variable



Slide 6 — f'' > 0 bends up, f'' = 0 is straight, f'' < 0 bends down.

$f'(t) = df/dt$ (slope), $f''(t) = \frac{d}{dt} f'(t)$ (change of slope). For $f = at^2 + bt + c$: $f'' = 2a$.

Summary: in one variable, $f''(t) \geq 0$ everywhere is the curvature test for a twice-differentiable convex function (A1).

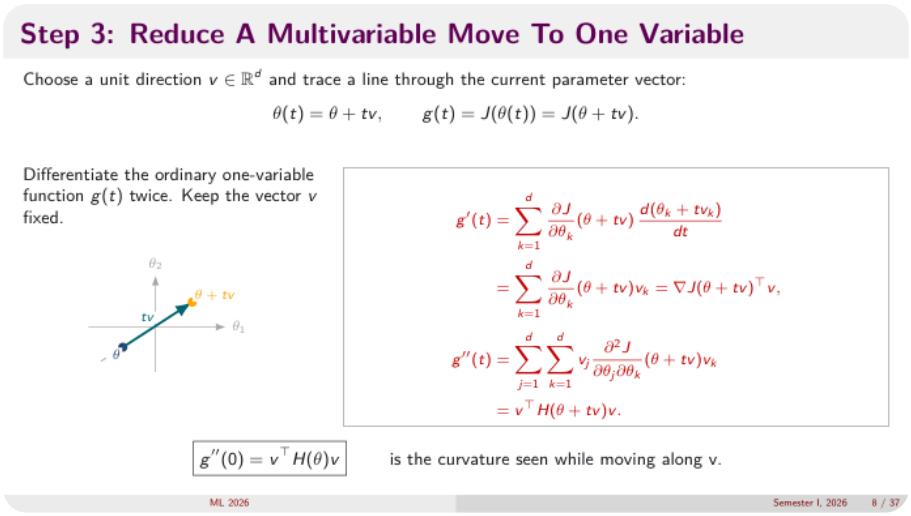
Slide 7 Step 2: Build The Hessian Matrix

The gradient stores the d first partials; the Hessian stores every second partial (A2). Diagonal = curvature along coordinate axes; off-diagonal = how changing one parameter alters the slope with respect to another.

Example: $J = 2\theta_1^2 + 2\theta_1\theta_2 + \theta_2^2 \Rightarrow \nabla J = (4\theta_1 + 2\theta_2,\ 2\theta_1 + 2\theta_2)$, $H = \begin{bmatrix} 4 & 2 \\ 2 & 2 \end{bmatrix}$. Every entry is worked in the A2 table.

Check at a point: at $\theta = (1, -1)$: $\nabla J = (4 - 2, 2 - 2) = (2, 0)$. So at that point J rises in θ_1 and is flat in θ_2.

Slide 8 Step 3: Reduce A Multivariable Move To One Variable



Slide 8 — Walk from θ along a fixed unit direction v; the height along the walk is a 1-D function g(t).

$\theta(t) = \theta + tv$, $g(t) = J(\theta + tv)$. Differentiate twice, keeping v fixed (A3):

$$g'(t) = \sum_k \frac{\partial J}{\partial \theta_k}(\theta + tv)\, v_k = \nabla J(\theta + tv)^\top v, \qquad g''(t) = \sum_j \sum_k v_j \frac{\partial^2 J}{\partial \theta_j \partial \theta_k}(\theta + tv)v_k = v^\top H(\theta + tv)v$$

$\boxed{g''(0) = v^\top H(\theta) v}$ is the curvature seen while moving along v.

Numbers: along $v = \frac{1}{\sqrt{2}}(1,1)$ from 0: $g(t) = J(t/\sqrt{2}, t/\sqrt{2}) = \frac{t^2}{2}(2+2+1) = 2.5t^2$, so $g'' = 5 = v^\top H v$ ✓.

Slide 9 Step 4: Taylor's Quadratic Approximation

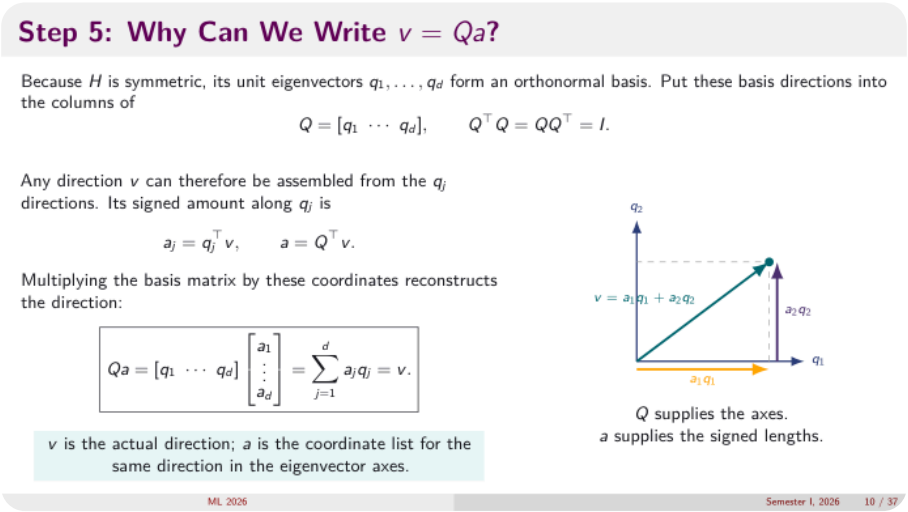
$g(t) \approx g(0) + tg'(0) + \frac{1}{2}t^2 g''(0) \Rightarrow$ with $\Delta = tv$:

$$J(\theta + \Delta) \approx J(\theta) + \nabla J(\theta)^\top \Delta + \frac{1}{2}\Delta^\top H(\theta)\Delta.$$

First-order information	Second-order information
$\nabla J^\top \Delta$ predicts the initial rise or fall along Δ	$\Delta^\top H \Delta$ measures how the slope changes along Δ. It's a **scalar**, not the vector $H\Delta$.

Summary: the Hessian appears *after* a direction turns the multivariable objective into a 1-D function (A4).

Slide 10 Step 5: Why Can We Write $v = Qa$?



Slide 10 — Q supplies the axes (eigenvectors), a supplies the signed lengths along them.

H symmetric ⇒ its unit eigenvectors $q_1..q_d$ form an orthonormal basis, $Q = [q_1 \cdots q_d]$, $Q^\top Q = QQ^\top = I$. Any direction:

$$a_j = q_j^\top v, \quad a = Q^\top v, \qquad Qa = \sum_j a_j q_j = v.$$

v is the actual direction; a is its coordinate list in eigen-axes. (Same "change of basis" as L5 §A5.)

Numbers: $H = \begin{bmatrix} 4 & 2 \\ 2 & 2 \end{bmatrix}$ has $q_{\max} \approx (0.851, 0.526)$ ($\lambda = 5.236$), $q_{\min} \approx (-0.526, 0.851)$ ($\lambda = 0.764$). For $v = (1, 0)$: $a = (0.851, -0.526)$, and $a_1^2 + a_2^2 = 0.724 + 0.276 = 1$ ✓.

Slide 11 Step 6: Eigenvalues Test Every Direction

$$v^\top H v = (Qa)^\top (Q\Lambda Q^\top)(Qa) = a^\top \Lambda a = \sum_{j=1}^{d} \lambda_j a_j^2, \qquad \sum_j a_j^2 = 1.$$

Directional curvature = **weighted average of eigenvalues** with weights a_j^2 (A5). Therefore

$$v^\top H v \geq 0 \; \forall v \iff \lambda_j \geq 0 \; \forall j.$$

Check slide 10's numbers: $v = (1, 0)$: $5.236(0.724) + 0.764(0.276) = 3.791 + 0.211 = 4.00 = H_{11}$ ✓.

Slide 12 Convex Can Still Have Many Minimizers

(Slide image in §A6.) $H \succeq 0 \iff \lambda_j \geq 0$.

Convex	Strictly convex
$\lambda_j \geq 0$: zero eigenvalues and flat directions allowed	$\lambda_j > 0$: every direction curves up ⇒ **at most one** minimiser

Example: $J(\theta_1, \theta_2) = \theta_1^2$. $H = \begin{bmatrix} 2 & 0 \\ 0 & 0 \end{bmatrix}$, $\lambda = 2, 0$. Every point on the line $\theta_1 = 0$ gives $J = 0$: infinitely many minimisers. Side view at any fixed θ_2: the same parabola.

Summary: a zero eigenvalue creates a flat direction: convex, but a whole line can minimise it. (Exactly what happens with duplicate predictors in L5/L7.)

Slide 13 The Bridge: Reuse The Curvature Test, Do Not Restart

	Linear regression	Logistic regression today
outcome	continuous y	binary $y \in \{0, 1\}$
probability model	Gaussian	Bernoulli
mean model	$X\theta$	$p = \sigma(X\theta)$
loss	squared error	to be derived (slide 25)
gradient	$X^\top(X\theta - y)$	to be derived (slide 30)
Hessian	$X^\top X$	to be derived (slide 32)
curvature test	$v^\top H v \geq 0$	**exactly the same test** (slide 33)

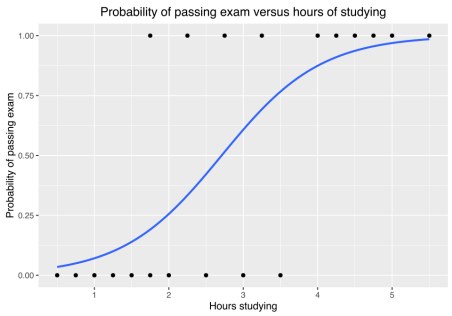
Question: what must change when the observation is binary, and which optimisation ideas stay? → The probability model and therefore the loss change; gradient/Hessian/convexity logic stays.

Slide 14 Running Example: Will This Road Segment Become Impassable?

For each road segment and forecast window: $x_i = (1,\ \text{rainfall},\ \text{slope},\ \text{drainage deficit})^\top$, $y_i \in \{0, 1\}$.

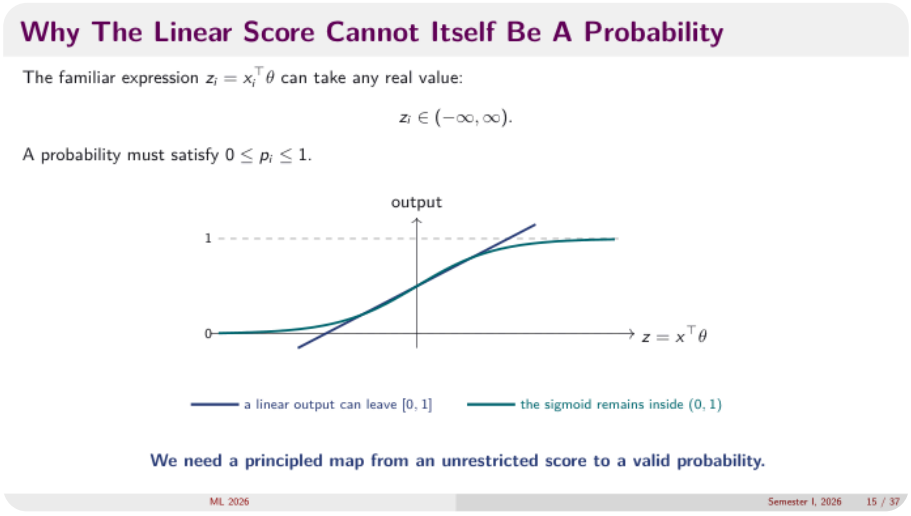
observed features x_i → linear score $z_i = x_i^\top \theta$ → probability p_i → operational action (close / monitor / open).

Careful. The model estimates a **probability**. The authority chooses an **action** using a threshold and the relative costs of missed closures and false alarms (slide 35, L9).



A real-world logistic fit: probability of passing an exam vs hours studied; same shape as the road model with one feature. (Wikimedia Commons: Exam_pass_logistic_curve.svg, CC BY-SA 4.0)

Slide 15 Why The Linear Score Cannot Itself Be A Probability

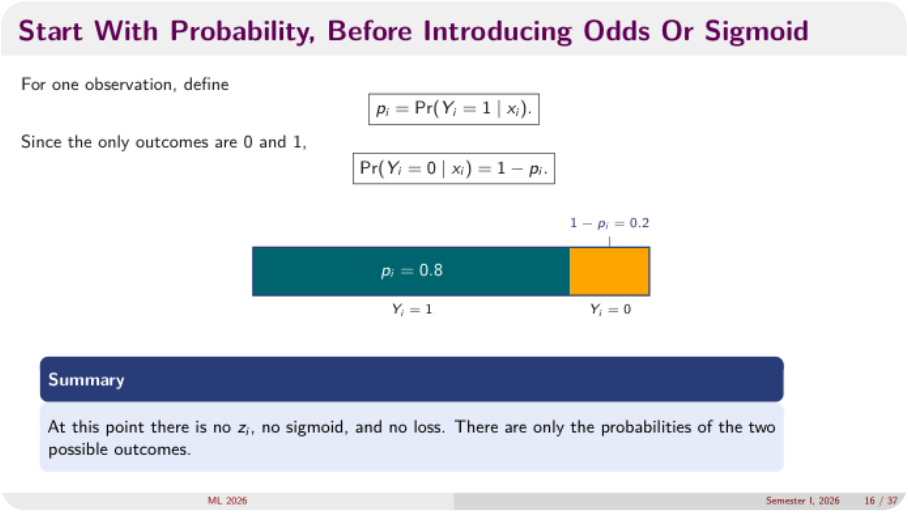


Slide 15 — A straight line leaves [0, 1]; the sigmoid stays inside (0, 1).

$z_i = x_i^\top \theta \in (-\infty, \infty)$, but $0 \le p_i \le 1$. Using z directly as a probability (the "linear probability model") gives impossible values: toy data with the least-squares line $\hat{p} = 0.0 + 0.2x$ would predict 1.2 at $x = 6$.

We need a principled map from an unrestricted score to a valid probability → slides 16–20.

Slide 16 Start With Probability, Before Introducing Odds Or Sigmoid



Slide 16 — Two outcomes, two probabilities that add to 1.

$p_i = \Pr(Y_i = 1 \mid x_i)$; since the only outcomes are 0 and 1, $\Pr(Y_i = 0 \mid x_i) = 1 - p_i$. Example: $p_i = 0.8$, $1 - p_i = 0.2$.

> **Summary:** at this point there's no z_i, no sigmoid, no loss. Only the probabilities of the two outcomes.

Slide 17 Probability To Odds: Compare Event With Non-Event

$$\text{odds}_i = \frac{\Pr(Y_i = 1 \mid x_i)}{\Pr(Y_i = 0 \mid x_i)} = \frac{p_i}{1 - p_i}. \qquad p_i = 0.8: \ \text{odds} = \frac{0.8}{0.2} = 4.$$

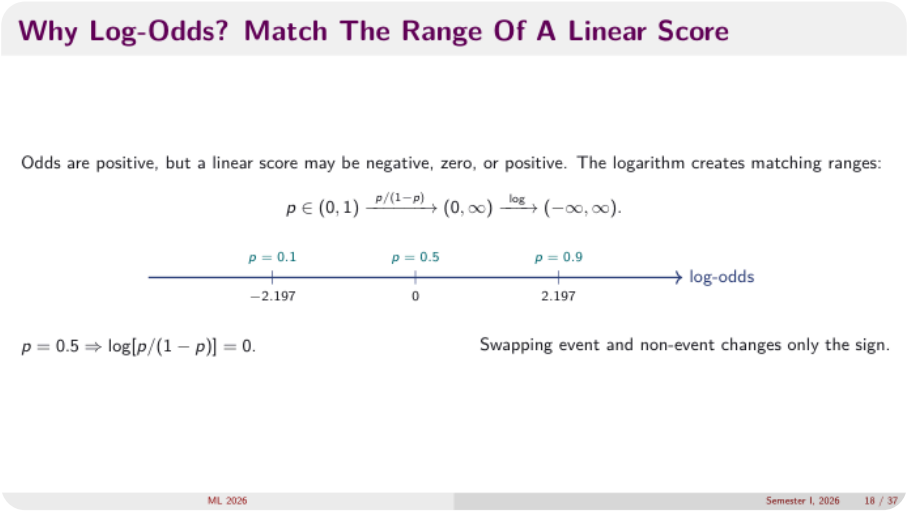
Probability 0.8 = 80 events per 100 comparable cases. Odds 4 = **four events for every one non-event**.

Back from odds: $p = \dfrac{\text{odds}}{1 + \text{odds}}$; odds 4 → $4/5 = 0.8$ ✓.

> **Summary:** probability compares events with all cases; odds compare events with non-events. Same information, different scale.

Slide 18

Why Log-Odds? Match The Range Of A Linear Score



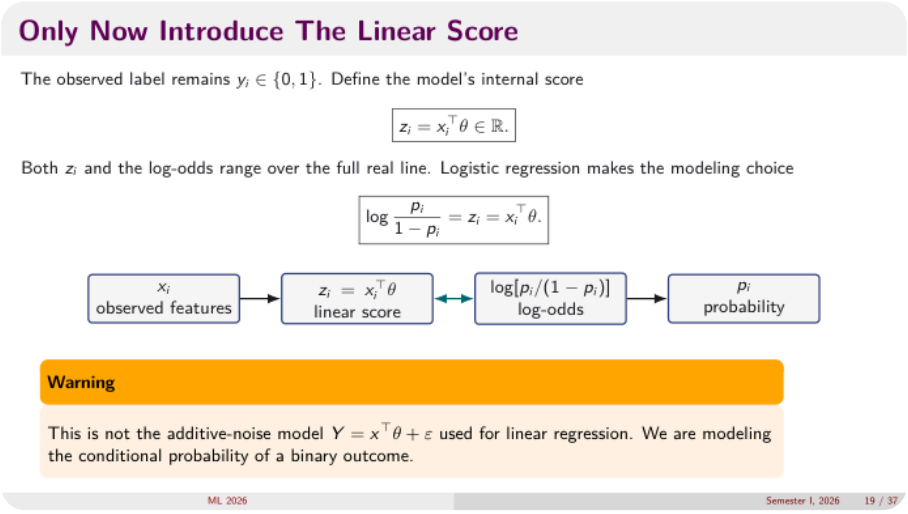
Slide 18 — log-odds of 0.1, 0.5, 0.9 are −2.197, 0, 2.197.

$$p \in (0,1) \xrightarrow{p/(1-p)} (0,\infty) \xrightarrow{\log} (-\infty,\infty).$$

- $p = 0.5 \Rightarrow \log 1 = 0.$
- $p = 0.9 \Rightarrow \log 9 = 2.197$; $p = 0.1 \Rightarrow \log \frac{1}{9} = -2.197.$
- Swapping event and non-event only flips the sign.

Now the log-odds and the score live on the **same range**, so we may set them equal.

Slide 19 Only Now Introduce The Linear Score



Slide 19 — features → score → log-odds → probability.

Labels stay $y_i \in \{0, 1\}$. Internal score $z_i = x_i^\top \theta \in \mathbb{R}$. **Modelling choice of logistic regression:**

$$\log \frac{p_i}{1 - p_i} = z_i = x_i^\top \theta.$$

Careful. This is **not** the additive-noise model $Y = x^\top \theta + \varepsilon$ of linear regression. We model the **conditional probability** of a binary outcome; there's no "noise term" added to y.

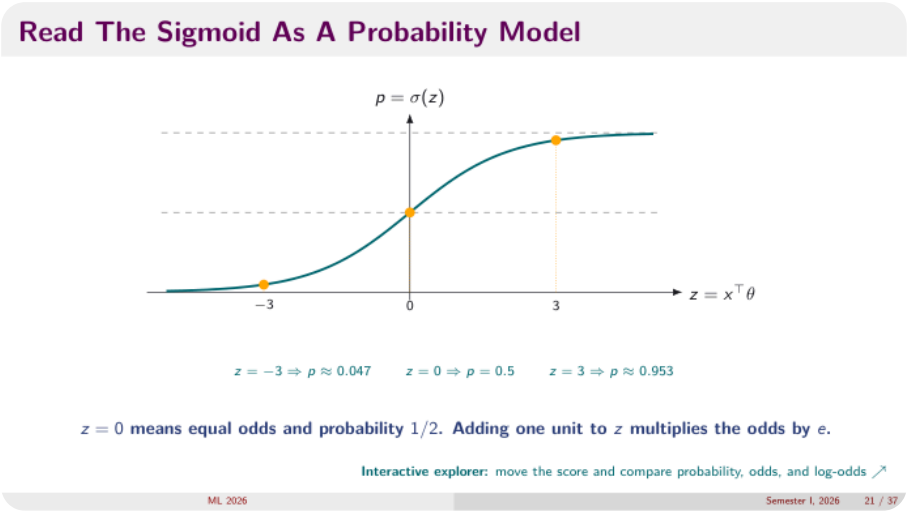
Toy: road 6 ($x = 5$) at $\hat{\theta}$: $z = -3.035 + 6.070 = 3.035$ = log-odds; odds $= e^{3.035} = 20.8$; $p = 20.8/21.8 = 0.954$.

Slide 20 Solve The Log-Odds Equation: The Sigmoid Appears

Line	Step
$\log \frac{p_i}{1-p_i} = z_i$	start
$\frac{p_i}{1-p_i} = e^{z_i}$	exponentiate
$p_i = e^{z_i}(1-p_i)$	multiply by $1-p_i$
$p_i = e^{z_i} - e^{z_i}p_i$	expand
$p_i(1+e^{z_i}) = e^{z_i}$	collect p_i
$p_i = \frac{e^{z_i}}{1+e^{z_i}} = \frac{1}{1+e^{-z_i}} = \sigma(z_i)$	divide; then top and bottom by e^{z_i}

Summary: the sigmoid is the inverse of the logit, not an arbitrary decorative function.

Slide 21 Read The Sigmoid As A Probability Model



Slide 21 — z = −3, 0, 3 → p ≈ 0.047, 0.5, 0.953.

z	e^{-z}	$p = 1/(1+e^{-z})$	odds e^{z}
-3	20.09	0.047	0.050
0	1	0.5	1
1	0.368	0.731	2.718
3	0.0498	0.953	20.09

$z = 0$ means equal odds, $p = \frac{1}{2}$. **Adding one unit to z multiplies the odds by e.** (Compare row $z = 0$ and $z = 1$.) The curve is symmetric about $(0, \frac{1}{2})$ and steepest there.

Slide 22 One Bernoulli Formula Covers Both Possible Labels

$$\Pr(Y_i = y_i \mid x_i, \theta) = p_i^{y_i}(1 - p_i)^{1-y_i}.$$

- $y_i = 1$: $p_i^1(1 - p_i)^0 = p_i$ → rewards a **large** p_i.
- $y_i = 0$: $p_i^0(1 - p_i)^1 = 1 - p_i$ → rewards a **small** p_i.

Question: if an event occurs but the model said $p_i = 0.01$, small or large penalty? → **Large**: probability of what happened is 0.01, so $-\log 0.01 = 4.6$ (A9 table).

Slide 23 From Individual Probabilities To The Dataset Likelihood

Assuming conditional independence given features:

$$L(\theta) = \prod_{i=1}^{n} p_i^{y_i}(1 - p_i)^{1-y_i}.$$

Each factor is "the probability the model gave to what actually happened": $y_1 = 1$ contributes p_1, $y_2 = 0$ contributes $1 - p_2$, ...

Careful. Conditional independence is **across observations** after conditioning on their inputs, not a claim that predictor columns are independent.

Toy at $\hat{\theta}$: $L = (1 - 0.046)(1 - 0.139)(0.353)(1 - 0.647)(0.861)(0.954) = 0.954 \cdot 0.861 \cdot 0.353 \cdot 0.353 \cdot 0.861 \cdot 0.954 = 0.0839$. Road 3 and road 4 (the "surprising" labels) contribute the two small factors.

Slide 24 Take Logs: Product Becomes A Sum

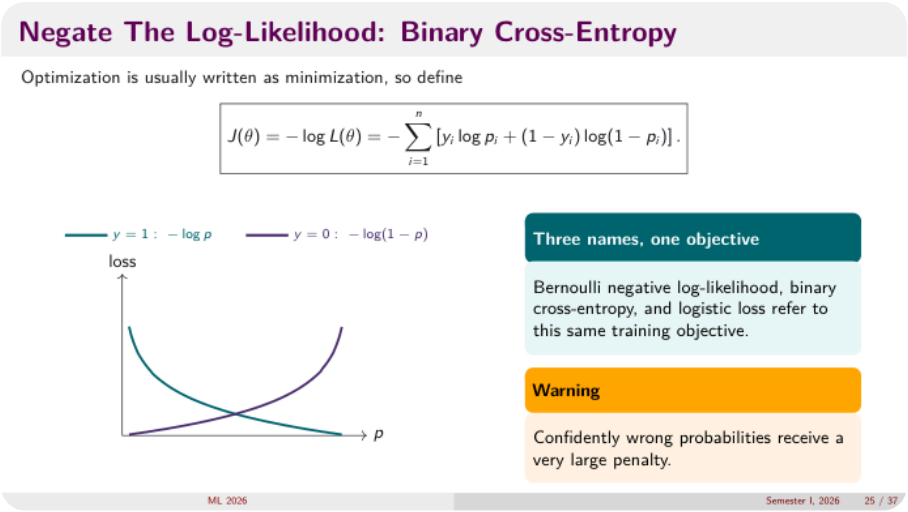
$$\log L(\theta) = \log \prod_i p_i^{y_i}(1-p_i)^{1-y_i} = \sum_i \log\left[p_i^{y_i}(1-p_i)^{1-y_i}\right] = \sum_i \left[y_i \log p_i + (1-y_i)\log(1-p_i)\right].$$

Rules used: $\log \prod a_i = \sum \log a_i$, $\log(a^b) = b \log a$, $\log(ab) = \log a + \log b$.

Toy: $\log 0.0839 = -2.478$ ✓ (matches $J(\hat{\theta}) = 2.478$ on the next slide).

Maximum likelihood chooses θ that makes the observed labels collectively most probable.

Slide 25 Negate The Log-Likelihood: Binary Cross-Entropy



Slide 25 — –log p (y = 1) and –log(1 – p) (y = 0): confidently wrong means a very large penalty.

$$J(\theta) = -\log L(\theta) = -\sum_{i=1}^{n} \left[y_i \log p_i + (1-y_i)\log(1-p_i)\right].$$

Three names, one objective: Bernoulli negative log-likelihood, binary cross-entropy, logistic loss.

Careful. Confidently wrong probabilities receive a very large penalty ($p \to 0$ when $y = 1 \Rightarrow -\log p \to \infty$).

Toy at $\theta = 0$: all $p_i = \frac{1}{2}$, so $J = 6 \log 2 = 4.159$. At $\hat{\theta}$: $J = 2.478$. Training lowered it by 1.68.

Slide 26 Substitute The Sigmoid: Compute The Log Terms First

With $p_i = e^{z_i}/(1+e^{z_i})$:

$$1 - p_i = \frac{1 + e^{z_i} - e^{z_i}}{1 + e^{z_i}} = \frac{1}{1 + e^{z_i}},$$

$$\log p_i = \log e^{z_i} - \log(1 + e^{z_i}) = z_i - \log(1 + e^{z_i}),$$

$$\log(1 - p_i) = \log 1 - \log(1 + e^{z_i}) = -\log(1 + e^{z_i}).$$

Slide 27 One-Observation Loss: Simplify Without Skipping A Line

$\ell_i = -y_i \log p_i - (1 - y_i)\log(1 - p_i)$. Insert slide 26:

$$\begin{aligned}
\ell_i &= -y_i\big[z_i - \log(1 + e^{z_i})\big] - (1 - y_i)\big[-\log(1 + e^{z_i})\big] \\
&= -y_i z_i + y_i \log(1 + e^{z_i}) + (1 - y_i)\log(1 + e^{z_i}) \\
&= -y_i z_i + \big[y_i + 1 - y_i\big]\log(1 + e^{z_i}) \\
&= \boxed{\log(1 + e^{z_i}) - y_i z_i}, \qquad z_i = x_i^\top \theta.
\end{aligned}$$

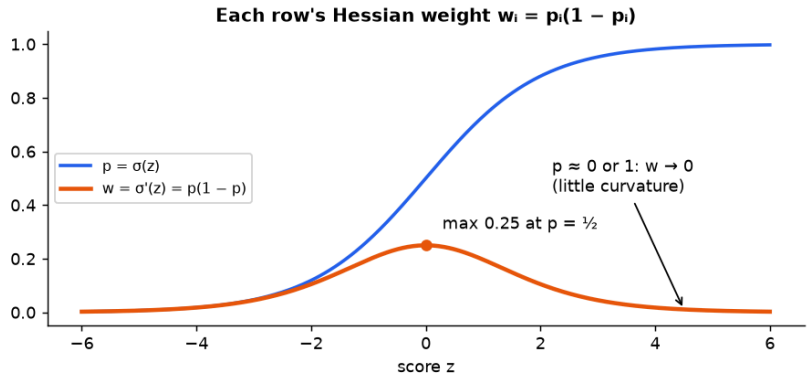
Check (road 3, $y = 1$, $z = -0.607$): $\log(1 + e^{-0.607}) + 0.607 = \log 1.545 + 0.607 = 0.435 + 0.607 = 1.042$; and $-\log p_3 = -\log 0.353 = 1.041$ ✓.

Slide 28 Check Both Labels: Does The Compact Formula Behave Correctly?

$y_i = 1$	$y_i = 0$
$\ell_i = \log(1 + e^{z_i}) - z_i = \log \frac{1+e^{z_i}}{e^{z_i}} = \log(1 + e^{-z_i})$	$\ell_i = \log(1 + e^{z_i})$
large **positive** score → small loss	large **negative** score → small loss

At $z = 0$ both are $\log 2 = 0.693$. **The log-odds model supplies p_i; the Bernoulli likelihood supplies the loss.**

Slide 29 Differentiate The Sigmoid Once, Then Reuse The Result



p and its slope p(1 – p): largest (0.25) at p = ½, near zero when p is close to 0 or 1.

$\sigma(z) = (1+e^{-z})^{-1}$. Chain rule (outer: $u^{-1} \to -u^{-2}$; inner: $\frac{d}{dz}(1+e^{-z}) = -e^{-z}$):

$$\sigma'(z) = -(1+e^{-z})^{-2} \cdot (-e^{-z}) = \frac{e^{-z}}{(1+e^{-z})^2} = \underbrace{\frac{1}{1+e^{-z}}}_{\sigma(z)} \cdot \underbrace{\frac{e^{-z}}{1+e^{-z}}}_{1-\sigma(z)} = \sigma(z)\big[1-\sigma(z)\big].$$

Therefore $\dfrac{dp_i}{dz_i} = p_i(1-p_i)$. Largest ($\frac{1}{4}$) at $p = \frac{1}{2}$, tends to 0 near 0 or 1.

Numbers: $p = 0.5 \to 0.25$; $p = 0.861 \to 0.120$; $p = 0.954 \to 0.044$.

Slide 30 Derive The Gradient: Score Residual Times Features

Use $\ell_i = \log(1+e^{z_i}) - y_i z_i$, $z_i = x_i^\top \theta$ (A10).

$$\frac{d\ell_i}{dz_i} = \frac{e^{z_i}}{1+e^{z_i}} - y_i = p_i - y_i, \qquad \nabla_\theta z_i = x_i,$$

$$\nabla_\theta \ell_i = (p_i - y_i)x_i, \qquad \boxed{\nabla J(\theta) = \sum_i (p_i - y_i)x_i = X^\top(p-y)}.$$

Same shape as linear regression's $X^\top(X\theta - y)$: **(prediction – truth), weighted by features**. Only "prediction" changed from $X\theta$ to $\sigma(X\theta)$.

Toy at $\theta = 0$: $p - y = (0.5, 0.5, -0.5, 0.5, -0.5, -0.5)$.

- intercept component: $\sum(p_i - y_i) = 0$.
- rainfall component: $0(0.5) + 1(0.5) + 2(-0.5) + 3(0.5) + 4(-0.5) + 5(-0.5) = -3.5$.

$\nabla J(0) = (0, -3.5)$: increase θ_1 (more rain ⇒ more risk) ✓.

At the optimum: $\nabla J = 0$ gives $\sum_i p_i = \sum_i y_i$ (intercept row): the fitted probabilities **add up to the observed number of events**. Toy: $0.046 + 0.139 + 0.353 + 0.647 + 0.861 + 0.954 = 3.000 = 3$ events ✓.

Slide 31 Derive The Hessian For One Observation

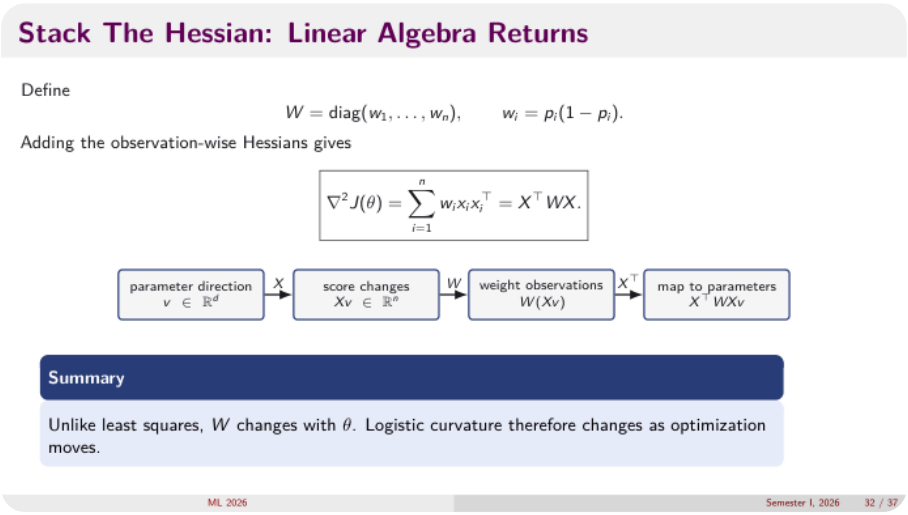
Differentiate $\nabla_\theta \ell_i = (p_i - y_i)x_i$ again. y_i and x_i are fixed data; only p_i depends on θ:

$$\nabla^2_\theta \ell_i = x_i(\nabla_\theta p_i)^\top, \qquad \nabla_\theta p_i = \frac{dp_i}{dz_i}\nabla_\theta z_i = p_i(1 - p_i)x_i,$$

$$\boxed{\nabla^2_\theta \ell_i = p_i(1 - p_i)\, x_i x_i^\top}.$$

The scalar $p_i(1 - p_i)$ decides how much curvature row i adds. Rows the model is **sure** about ($p \approx 0$ or 1) add almost none.

Slide 32 Stack The Hessian: Linear Algebra Returns



Slide 32 — v → Xv (score changes) → W(Xv) (weight each row) → $X^\top WXv$ (back to parameter space).

$W = \mathrm{diag}(w_1..w_n)$, $w_i = p_i(1-p_i)$. Add the per-row Hessians (A11):

$$\nabla^2 J(\theta) = \sum_i w_i x_i x_i^\top = X^\top W X.$$

Toy at $\theta = 0$: $w_i = 0.25$ for all, so $H = 0.25 X^\top X = 0.25 \begin{bmatrix} 6 & 15 \\ 15 & 55 \end{bmatrix} = \begin{bmatrix} 1.5 & 3.75 \\ 3.75 & 13.75 \end{bmatrix}$. ($X^\top X$: $n = 6$, $\sum x = 15$, $\sum x^2 = 0+1+4+9+16+25 = 55$.)

At $\hat{\theta}$: $w = (0.044, 0.120, 0.228, 0.228, 0.120, 0.044)$, $H = \begin{bmatrix} 0.784 & 1.960 \\ 1.960 & 6.100 \end{bmatrix}$: smaller, because confident rows contribute less.

> **Summary:** unlike least squares, W changes with θ, so **logistic curvature changes as optimisation moves**.

Slide 33 Apply The Opening Theorem: Logistic Loss Is Convex

For any direction v:

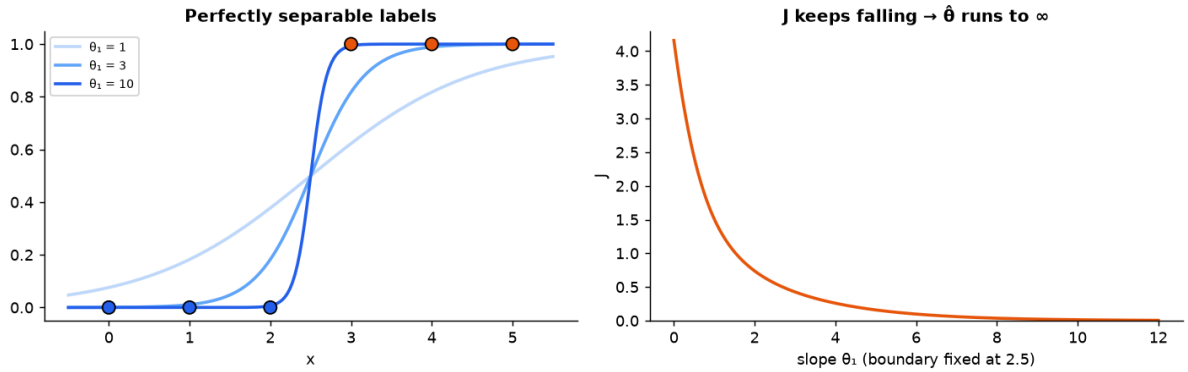
$$v^\top \nabla^2 J(\theta) v = v^\top X^\top W X v = (Xv)^\top W (Xv) = \sum_{i=1}^{n} p_i(1-p_i)(x_i^\top v)^2 \ \geq 0.$$

Each weight is positive ($0 < p_i < 1$), each square non-negative ⇒ $\nabla^2 J \succeq 0$ **for every** θ ⇒ J is convex (A5). Any point with $\nabla J = 0$ is a global minimum.

When is it *strictly* positive? $v^\top H v = 0$ needs $x_i^\top v = 0$ for every i, i.e. $Xv = 0$. If X has **full column rank**, that forces $v = 0$ ⇒ $H \succ 0$ ⇒ strictly convex ⇒ at most one minimiser.

Convexity is **not enough** for a finite, unique estimate:

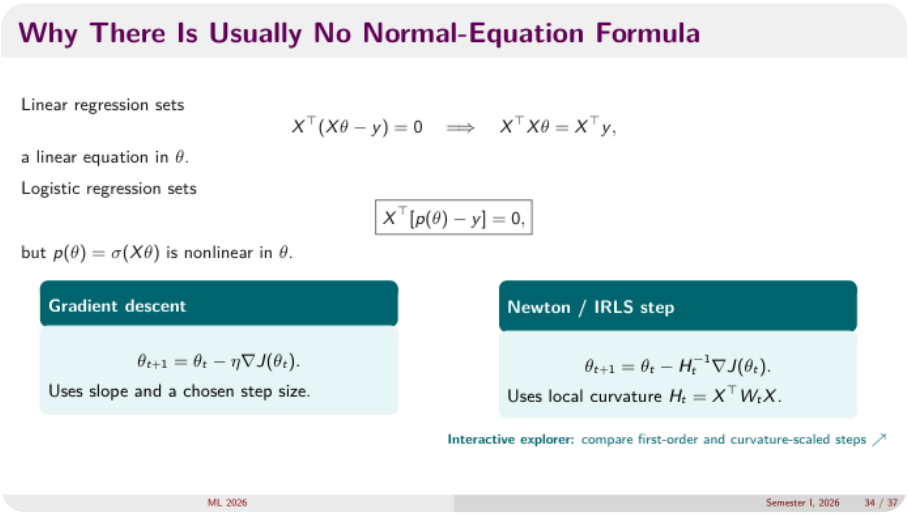
Problem	What happens
rank deficiency (duplicate/collinear columns)	flat direction $Xv = 0$ ⇒ a whole line of minimisers (slide 12)
complete separation (a line splits the classes perfectly)	J keeps decreasing as $\|\theta\| \to \infty$ ⇒ no finite minimiser



Perfectly separable toy labels: steeper and steeper sigmoids keep lowering J toward 0, so the slope θ_1 runs off to infinity.

(With labels $0, 0, 0, 1, 1, 1$ every p_i can be pushed toward the right side by making the slope bigger, so no finite θ is best. Fix: regularisation, L7 slide 33. Our running toy has roads 3 and 4 "crossing", so it is not separable and $\hat{\theta}$ is finite.)

Slide 34 Why There Is Usually No Normal-Equation Formula



Slide 34 — Gradient descent uses slope + step size; Newton/IRLS uses local curvature $H_t = X^\top W_t X$.

Linear regression	Logistic regression
$X^\top(X\theta - y) = 0 \Rightarrow X^\top X\theta = X^\top y$: **linear** in θ	$X^\top[p(\theta) - y] = 0$ with $p = \sigma(X\theta)$: **nonlinear** in θ

So we iterate (A12):

- **Gradient descent:** $\theta_{t+1} = \theta_t - \eta\nabla J(\theta_t)$.

- **Newton / IRLS:** $\theta_{t+1} = \theta_t - H_t^{-1}\nabla J(\theta_t)$, $H_t = X^\top W_t X$.

One Newton step by hand (toy, from $\theta_0 = 0$). $g = (0, -3.5)$, $H = \begin{bmatrix} 1.5 & 3.75 \\ 3.75 & 13.75 \end{bmatrix}$, $\det H = 20.625 - 14.0625 = 6.5625$.

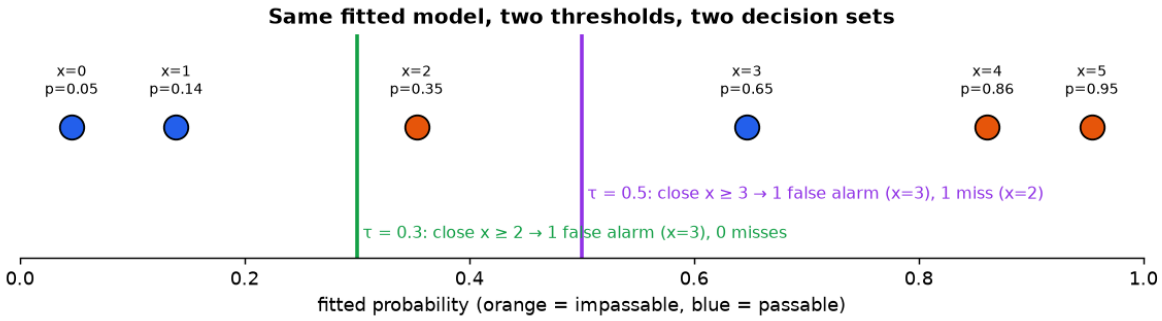
$$H^{-1}g = \frac{1}{6.5625}\begin{bmatrix} 13.75 & -3.75 \\ -3.75 & 1.5 \end{bmatrix}\begin{bmatrix} 0 \\ -3.5 \end{bmatrix} = \frac{1}{6.5625}\begin{bmatrix} 13.125 \\ -5.25 \end{bmatrix} = \begin{bmatrix} 2 \\ -0.8 \end{bmatrix}, \qquad \theta_1 = 0 - (2, -0.8) = (-$$

step	θ	J	$\|\nabla J\|$
0	$(0, 0)$	4.159	3.5
1	$(-2.000, 0.800)$	2.607	0.69
2	$(-2.774, 1.110)$	2.485	0.14
3	$(-3.015, 1.206)$	2.4780	0.0095
4	$(-3.035, 1.214)$	2.4780	0.0001

The gradient shrinks ~×5, ×5, ×14, ×95: Newton's characteristic quadratic speed-up near the optimum. **Meaning of $\hat{\theta}$:** each extra unit of rain multiplies the odds of impassability by $e^{1.214} = 3.37$; $p = \frac{1}{2}$ at $x = 3.035/1.214 = 2.5$.

Slide 35

A Probability Is Not Yet A Decision



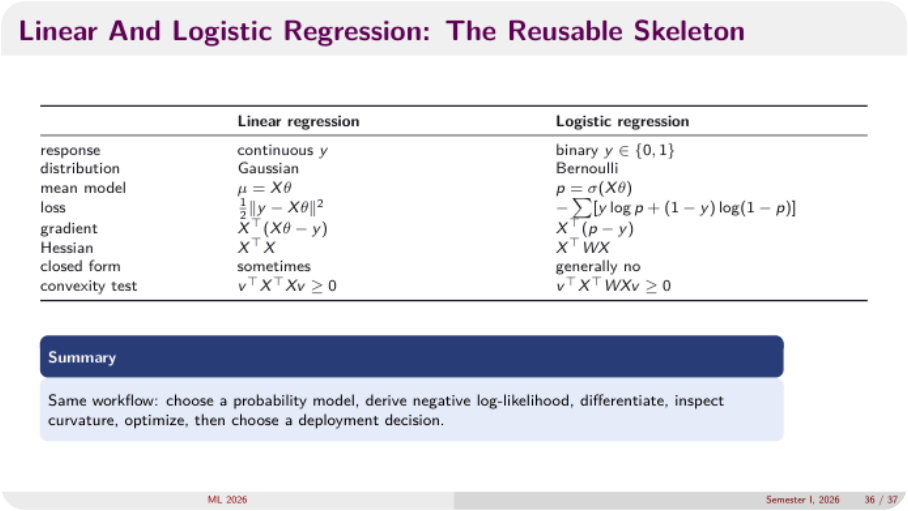
The six fitted probabilities with two thresholds: τ = 0.5 closes roads 4–6 (one false alarm, one miss); τ = 0.3 also closes road 3 (one false alarm, no miss).

After fitting, an operational rule uses a threshold τ: $\hat{y}_i = \mathbb{I}[p_i \geq \tau]$.

Lower τ closes more roads	Higher τ closes fewer roads
fewer missed impassable segments	fewer unnecessary closures
more unnecessary closures	more missed impassable segments

Careful. Maximum likelihood estimates **probabilities**. The threshold expresses a **decision cost** and can be changed **without refitting** the probability model. (Full treatment: L9–L10.)

Slide 36 Linear And Logistic Regression: The Reusable Skeleton



Slide 36 — The two models side by side.

	Linear regression	Logistic regression
response	continuous y	binary $y \in \{0, 1\}$
distribution	Gaussian	Bernoulli
mean model	$\mu = X\theta$	$p = \sigma(X\theta)$
loss	$\frac{1}{2}\|y - X\theta\|^2$	$-\sum[y \log p + (1-y)\log(1-p)]$
gradient	$X^\top(X\theta - y)$	$X^\top(p - y)$
Hessian	$X^\top X$ (constant)	$X^\top WX$ (changes with θ)
closed form	sometimes	generally no
convexity test	$v^\top X^\top Xv \geq 0$	$v^\top X^\top WXv \geq 0$

Summary: same workflow: choose a probability model → negative log-likelihood → differentiate → inspect curvature → optimise → choose a deployment decision.

Slide 37 What You Should Be Able To Re-Derive

1. $v^\top H v = \sum_j \lambda_j a_j^2$ and the eigenvalue test for convexity (slides 8–11);
2. $p \to p/(1-p) \to \log[p/(1-p)] = x^\top \theta \to p = \sigma(x^\top \theta)$ (slides 16–20);
3. Bernoulli likelihood → binary cross-entropy (slides 22–25);
4. $\nabla J = X^\top (p - y)$ and $\nabla^2 J = X^\top W X$ (slides 29–32);
5. $v^\top X^\top W X v \geq 0$, hence convexity (slide 33);
6. why **convexity, uniqueness, finite coefficients, and a decision threshold** are four different questions (A6, slides 33, 35).

Part C Practice

1. For $J(\theta_1, \theta_2) = \theta_1^2 + 4\theta_1\theta_2 + \theta_2^2$, find H and its eigenvalues. Is J convex?
2. For $H = \begin{bmatrix} 4 & 2 \\ 2 & 2 \end{bmatrix}$, compute $v^\top H v$ for $v = \frac{1}{5}(3, 4)$ and check it lies between the eigenvalues.
3. Convert $p = 0.25$ to odds and to log-odds. What is the log-odds of $p = 0.75$?
4. Show $\sigma(-z) = 1 - \sigma(z)$.
5. A model has $z = -1 + 0.5x$. What is p at $x = 2$? By what factor do the odds change when x goes from 2 to 4?
6. Compute the binary cross-entropy for three examples: $(y, p) = (1, 0.9), (0, 0.2), (1, 0.3)$.
7. Using $\ell = \log(1 + e^z) - yz$, verify that $y = 0$, $z = 2$ gives the same value as $-\log(1 - \sigma(2))$.
8. Derive $\frac{d}{dz}\log(1 + e^z)$ and show it equals $\sigma(z)$.
9. For the toy data at $\theta = (-2, 0.8)$ (Newton step 1), the fitted $p = (0.119, 0.231, 0.401, 0.599, 0.769, 0.881)$. Compute ∇J.
10. Show $\sum_i w_i x_i x_i^\top = X^\top W X$ for $n = 2$, $d = 2$ by writing both sides out.
11. Give a dataset with one feature where logistic regression has no finite MLE. Why does the intercept-only model always have one (when both labels appear)?
12. Why can the probability threshold be changed after training, but not the loss?

1. $H = \begin{bmatrix} 2 & 4 \\ 4 & 2 \end{bmatrix}$, $\lambda = 2 \pm 4 = 6, -2$. A negative eigenvalue ⇒ **not convex** (it's a saddle: along $(1, -1)$ the function curves down).
2. $Hv = \frac{1}{5}(12 + 8,\ 6 + 8) = \frac{1}{5}(20, 14)$; $v^\top H v = \frac{1}{25}(3 \cdot 20 + 4 \cdot 14) = \frac{116}{25} = 4.64 \in [0.764, 5.236]$ ✓.
3. odds $= 0.25/0.75 = 1/3$; log-odds $= \log(1/3) = -1.099$. For $p = 0.75$: $+1.099$ (sign flip).
4. $\sigma(-z) = \frac{1}{1+e^z} = \frac{e^{-z}}{e^{-z}+1}$ and $1 - \sigma(z) = \frac{1+e^{-z}-1}{1+e^{-z}} = \frac{e^{-z}}{1+e^{-z}}$ ✓.
5. $z = 0 \Rightarrow p = 0.5$. Odds multiply by $e^{0.5 \cdot 2} = e \approx 2.72$.
6. $-\log 0.9 = 0.105$; $-\log 0.8 = 0.223$; $-\log 0.3 = 1.204$. Total 1.532.
7. $\log(1 + e^2) - 0 = \log 8.389 = 2.127$; $\sigma(2) = 0.8808$, $-\log 0.1192 = 2.127$ ✓.
8. $\frac{e^z}{1+e^z} = \sigma(z)$ (chain rule: $\frac{1}{1+e^z} \cdot e^z$).
9. $p - y = (0.119, 0.231, -0.599, 0.599, -0.231, -0.119)$. Intercept: sum $= 0.000$. Rainfall: $0 + 0.231 - 1.198 + 1.797 - 0.924 - 0.595 = -0.689$. $\nabla J \approx (0, -0.69)$ (matches the table: $\|\nabla J\| = 0.69$).

10. LHS entry $(j,k) = w_1 x_{1j} x_{1k} + w_2 x_{2j} x_{2k}$. RHS: $(WX)_{ik} = w_i x_{ik}$, so $(X^\top WX)_{jk} = \sum_i x_{ij} w_i x_{ik}$: same.
11. $x = (1,2,3,4)$, $y = (0,0,1,1)$: the threshold 2.5 separates perfectly; scaling θ up keeps lowering J. Intercept-only: $J(\theta_0) = n\log(1+e^{\theta_0}) - k\theta_0$ with $0 < k < n$ events; it → ∞ at both ends, so a finite minimum exists at $\sigma(\theta_0) = k/n$.
12. The loss defines **what probabilities are fitted** (training). The threshold only maps fixed probabilities to actions; changing it re-labels decisions without touching θ (slide 35, L9).

Next: Lecture 7: when predictors are correlated, the Hessian has tiny eigenvalues; **regularisation** fixes that.